Dimension and Rank (Workshop)

Geometric Algorithms

CAS CS 132

Objectives

- Discuss the **coordinate systems**
- Introduce the fundamental notion of **dimension** which quantifies how "large" a space is
- Relate the dimension of the **column space** and the **null space** of a matrix (rank–nullity thm)

Keywords

basis

column space

null space

coordinate system

change of basis

dimension

rank

rank-nullity theorem

invertible matrix theorem (extended)

How to: Finding a basis for the null space

Q. Given a $m \times n$ matrix A find a basis for $\text{Nul}(A)$

A.



1. Find a general form solution for $A\mathbf{x} = \mathbf{0}$
2. Write this solution as a linear combination of vectors *where the free variables are the weights*
3. The resulting vectors form a basis for $\text{Nul}(A)$

How To: Finding a basis for the column space

Q. Given a $m \times n$ matrix A, find a basis for $\text{Col}(A)$

A.

1. Find the pivot columns in an echelon form of A
2. The associated columns <u>in A</u> form a basis for $\text{Col}(A)$

workshop (part 1)

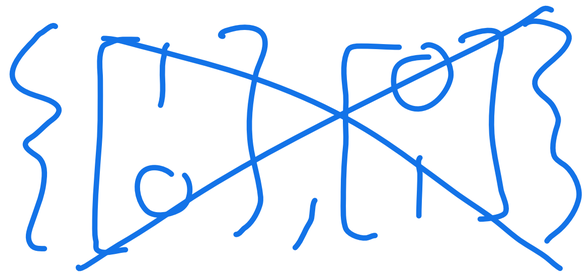
Q2

$$A = \begin{bmatrix} 1 & -4 & 3 & -3 \\ -2 & 8 & -6 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & -4 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

basis for $Col(A)$

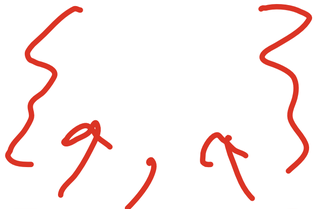
$\dim(Col(A)) = 2$

$$\left\{ \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \begin{bmatrix} -3 \\ 7 \end{bmatrix} \right\}$$



basis for $Nul(A)$

$\dim(Nul(A)) = 2$



$Ax = 0$

gen. sol'n

$x_1 = 4x_2 - 3x_3$

x_2 free

x_3 free

$x_4 = 0$

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = x_2 \begin{bmatrix} 4 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -3 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

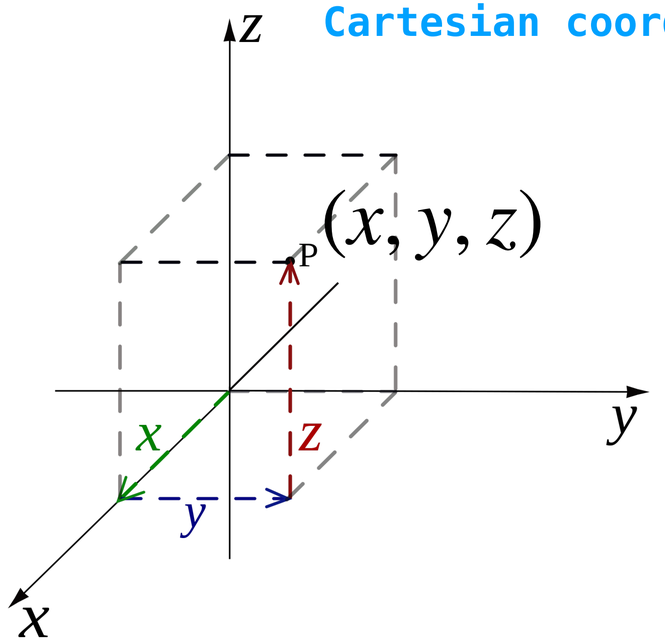
Coordinate Systems

At a High Level

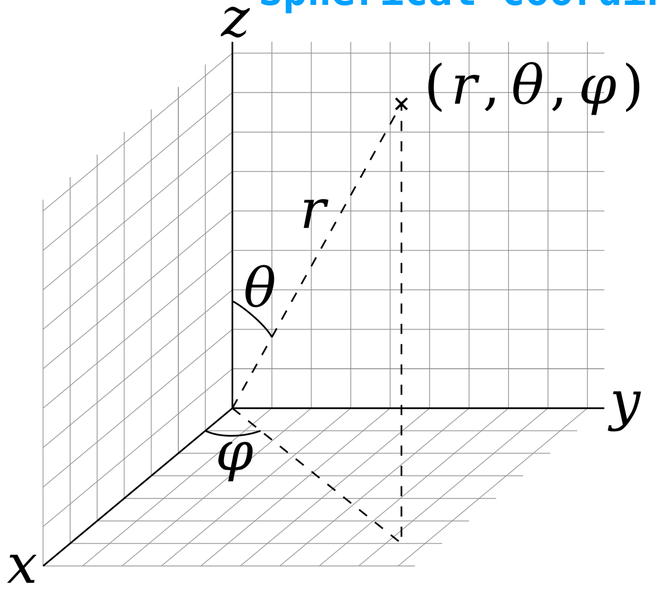
A coordinate system is a way of representing *positions* in terms of a *sequence of numbers*

Examples.

Cartesian coordinates



Spherical coordinates



Question*

Is $(2.3, 0.01, 5)$ *a spherical coordinate or a cartesian coordinate?*

This question is non-sensical

It's just a sequence of numbers. We need to be *told* if it should be interpreted in the **spherical** coordinate system or the **Cartesian** coordinate system

*Admittedly a trick question

Bases define Coordinate Systems

Given a basis $\mathscr{B}$ of a subspace H, there is **exactly one way** to write every vector in H as a linear combination of vectors in $\mathscr{B}$

Every basis provides a way to write down *coordinates* of a vector

And every time we write down a vector, we are **assuming a coordinate system**

Standard Basis

The standard basis defines the Cartesian coordinate system for $\mathbb{R}^n$

$$\begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} = 1\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 3\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + 2\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2 + \ldots + v_n\mathbf{e}_n$$

Column vectors are just weights for a linear combination of the standard basis

A Thought Experiment

Imagine doing this whole class from the beginning, but never saying *what vectors are*

(This is actually how we would do linear algebra if this were a math class)

Then one day, you get tired of talking about "abstract" vectors, you want to work with *numbers*

A Thought Experiment

Because we've learned everything up to now, we know that there is a basis $\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_n$ for the space $\mathbb{R}^n$

So given $\mathbf{v}$, if we know how to write it in terms of the basis, we can write...

$$\mathbf{v} = 2\mathbf{b}_1 + 3\mathbf{b}_2 + \ldots + (-0.1)\mathbf{b}_n \qquad [\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} 2 \\ 3 \\ \vdots \\ -0.1 \end{bmatrix}$$

and then choose those weights as a representation of v

But wait...

This *depends* on the choice of basis

If we started with $\mathbf{c}_1, \mathbf{c}_2, \ldots, \mathbf{c}_n$ then we would get some other representation.

$$[\mathbf{v}] = (-10)\mathbf{c}_1 + (4.3)\mathbf{c}_2 + \ldots + 0\mathbf{c}_n = \begin{bmatrix} -10 \\ 4.3 \\ \vdots \\ 0 \end{bmatrix}$$

Every basis defines a different coordinate system

Standard Basis

$$\begin{bmatrix}1\\3\\2\end{bmatrix} = -1\begin{bmatrix}-1\\0\\0\end{bmatrix} + 3\begin{bmatrix}0\\1\\0\end{bmatrix} + 2\begin{bmatrix}0\\0\\1\end{bmatrix}$$

The standard basis defines the Cartesian coordinate system for $\mathbb{R}^n$

$$[\vec{v}]_{\mathcal{B}} = \begin{bmatrix}-1\\3\\2\end{bmatrix} \leftarrow$$

$$\vec{v} = \begin{bmatrix}1\\3\\2\end{bmatrix}$$

$$\begin{bmatrix}v_1\\v_2\\\vdots\\v_n\end{bmatrix} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2 + \ldots + v_n\mathbf{e}_n$$

$\vec{e}_1, \vec{e}_2, \vec{e}_3$

$\mathcal{B} = \vec{e}_1, \vec{e}_2, \vec{e}_3$

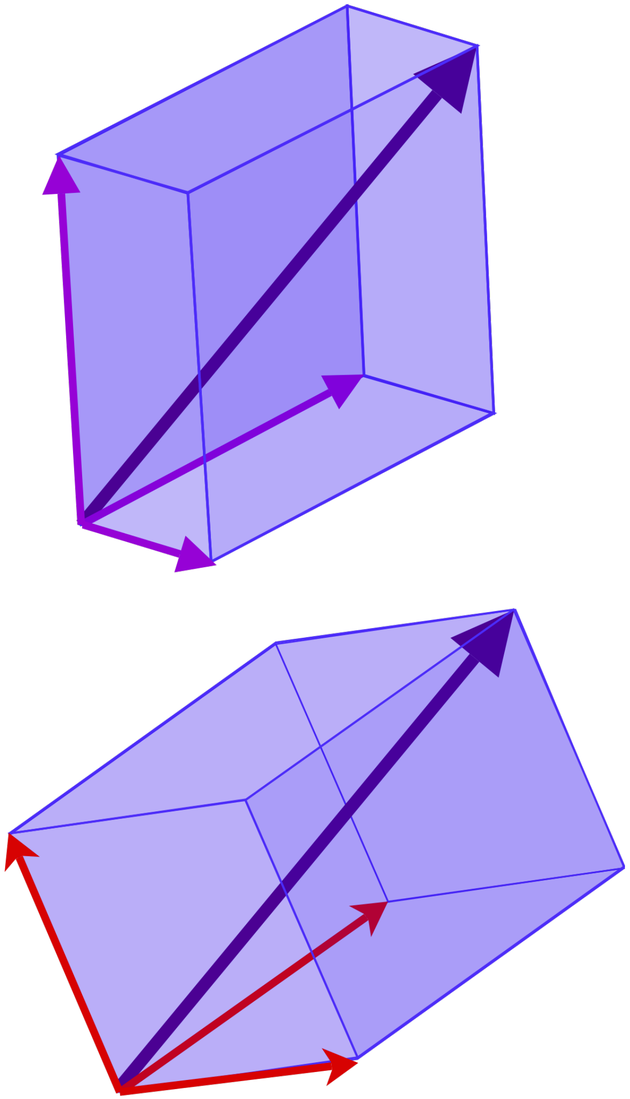
Column vectors are just weights for a linear combination of the standard basis

but we can also use different coordinate systems

How to think about this

Changing the coordinate system "warps space"

The question is: how do we represent a vector $\mathbf{v}$ in the warped space if we wanted it to "be in the same place"?



https://commons.wikimedia.org/wiki/File:3d_two_bases_same_vector.svg

ILA Demo

https://textbooks.math.gatech.edu/ila/bases-as-coord-systems.html

Coordinate Vectors

Let $\mathbf{v}$ be a vector in a subspace H of $\mathbb{R}^n$ and let $\mathscr{B} = \{\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_k\}$ be a basis of H where

$$\mathbf{v} = a_1\mathbf{b}_1 + a_2\mathbf{b}_2 + \ldots + a_k\mathbf{b}_k$$

Def. The **coordinate vector of $\mathbf{v}$ relative to $\mathscr{B}$** is

$$[\mathbf{v}]_{\mathscr{B}} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_k \end{bmatrix}$$

Coordinate Vectors and the Standard Basis

When we write down a vector $\mathbf{v}$ in $\mathbb{R}^n$, we're really writing down a coordinate vector **relative to the standard basis** $\mathscr{E}$

$$[\mathbf{v}]_{\mathscr{E}} = \mathbf{v}$$

How do we find coordinate vectors?

For an arbitrary basis $\mathscr{B}$, to determine $[\mathbf{v}]_{\mathscr{B}}$, we need to find weights $a_1, \ldots, a_k$ such that

$$a_1\mathbf{b}_1 + \ldots + a_k\mathbf{b}_k = \mathbf{v} \iff \begin{bmatrix} \vec{b}_1 & \cdots & \vec{b}_k \end{bmatrix} \begin{bmatrix} a_1 \\ \vdots \\ a_k \end{bmatrix} = \begin{bmatrix} \vec{v} \end{bmatrix}$$

This is just *solving a vector equation*

How To: Coordinate Vectors

Q. Find the coordinate vector for $\mathbf{v}$ in the subspace H relative to the basis $\mathbf{b}_1, \ldots, \mathbf{b}_k$

A. Solve the vector equation $x_1\mathbf{b}_1 + \ldots + x_k\mathbf{b}_k = \mathbf{v}$. A solution $(a_1, \ldots, a_k)$ means

$$[\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} a_1 \\ \vdots \\ a_k \end{bmatrix}$$

$$\begin{bmatrix} \vec{b}_1 & \cdots & \vec{b}_k \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_k \end{bmatrix} = \begin{bmatrix} \vec{v} \end{bmatrix}$$

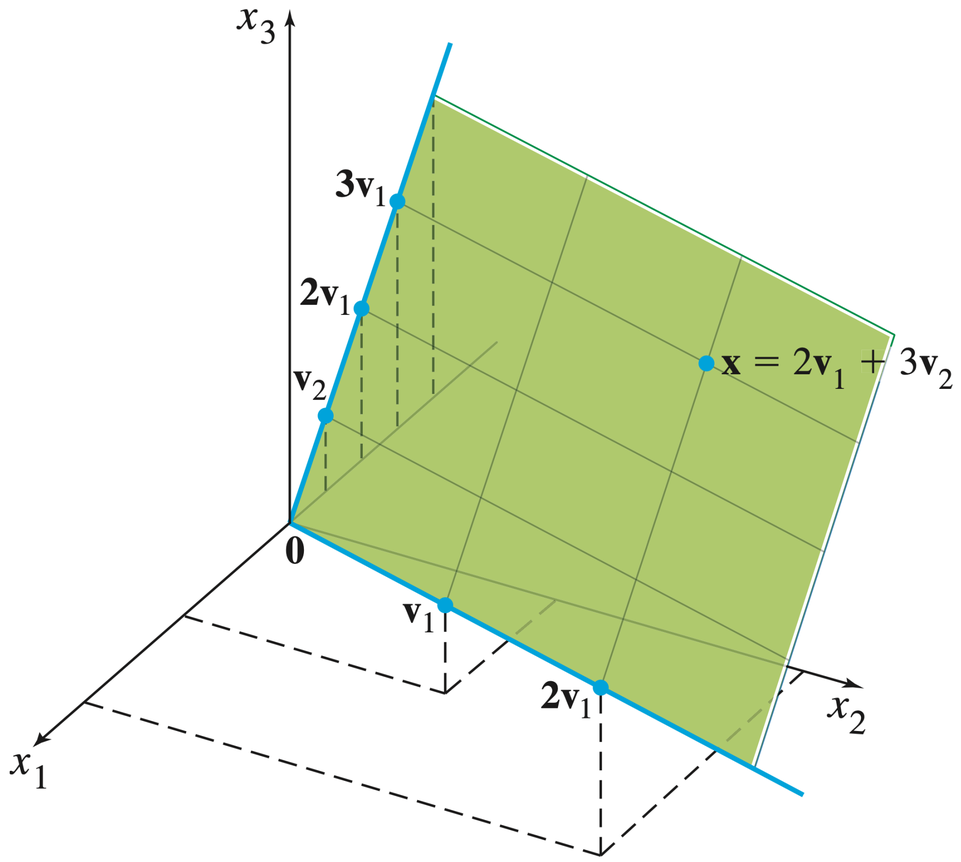
Coordinates and one-to-one correspondences

In the previous example $\mathbf{x} \mapsto [\mathbf{x}]_{\mathcal{B}}$ is a *one-to-one correspondence* from H to $\mathbb{R}^2$. This is also sometimes called an **isomorphism**

Isomorphic things "look and behave the same up to simple transformations

So $\text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$ is *isomorphic* to $\mathbb{R}^2$

This is a formal way of saying that $\text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$ is a "copy of $\mathbb{R}^2$"



Linear Algebra and its Applications (Lay, Lay, McDonald)

workshop (part 2)

Q4)

Solve $x_1\begin{bmatrix}3\\6\\2\end{bmatrix}+x_2\begin{bmatrix}-1\\0\\1\end{bmatrix}=\begin{bmatrix}3\\12\\7\end{bmatrix}\Leftrightarrow\begin{bmatrix}3&-1\\6&0\\2&1\end{bmatrix}\begin{bmatrix}x_1\\x_2\end{bmatrix}=\begin{bmatrix}3\\12\\7\end{bmatrix}$

ξ, $\beta=\beta$, $=\vec{v}$

$$\left[\begin{array}{cc|c}3&-1&3\\6&0&12\\2&1&7\end{array}\right]\sim\left[\begin{array}{cc|c}3&-1&3\\0&2&6\\2&1&7\end{array}\right]\sim\left[\begin{array}{cc|c}1&-2&-4\\0&2&6\\2&1&7\end{array}\right]$$

$$\sim\left[\begin{array}{cc|c}1&-2&-4\\0&2&6\\0&5&15\end{array}\right]\sim\left[\begin{array}{cc|c}1&-2&-4\\0&1&3\\0&0&0\end{array}\right]\sim\left[\begin{array}{cc|c}1&0&2\\0&1&3\\0&0&0\end{array}\right]$$

$$[v]_{\mathcal{B}}=\begin{bmatrix}x_1\\x_2\end{bmatrix}=\begin{bmatrix}2\\3\end{bmatrix}$$

Q5

$$\rightarrow [u]_{\mathcal{B}} = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$$

(23) $\left\{ \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$

$\vec{b}_1$ $\quad$ b_2

$$\rightarrow \vec{u} = 2\begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix} - 2\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 12 \\ 4 \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \\ -2 \end{bmatrix} = \begin{bmatrix} 8 \\ 12 \\ 2 \end{bmatrix}$$

Dimension and Rank

The Idea of Dimension

Thm. Every basis of a subspace H has *exactly the same number of vectors*

Any fewer, we wouldn't cover everything

Any more, we would have dependencies

This number is a measure of how "large" H is

Dimension

Def. The **dimension** of a subspace H of $\mathbb{R}^n$, written $\dim H$, is the *number* of vectors in *any* basis of H

We say H is k**–dimensional** if it has dimension k

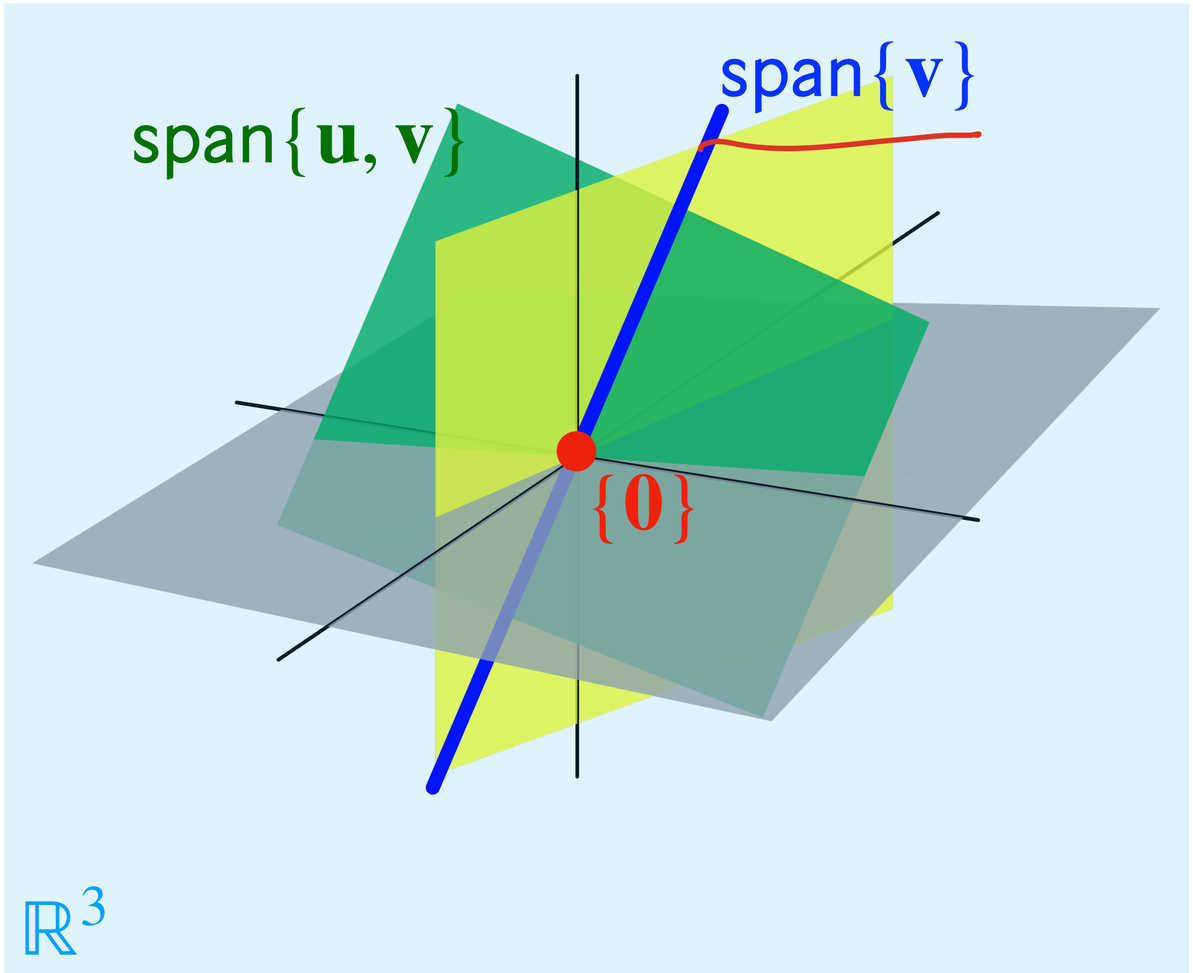
This should confirm our intuitions:

» a plane (through the origin) is a 2D subspace
» a line (through the origin) is a 1D subspace

Recall: Subspace in $\mathbb{R}^3$

There are only 4 kinds of subspaces of $\mathbb{R}^3$:

1. $\{\mathbf{0}\}$ just the origin
2. lines (through the origin)
3. planes (through the origin)
4. All of $\mathbb{R}^3$

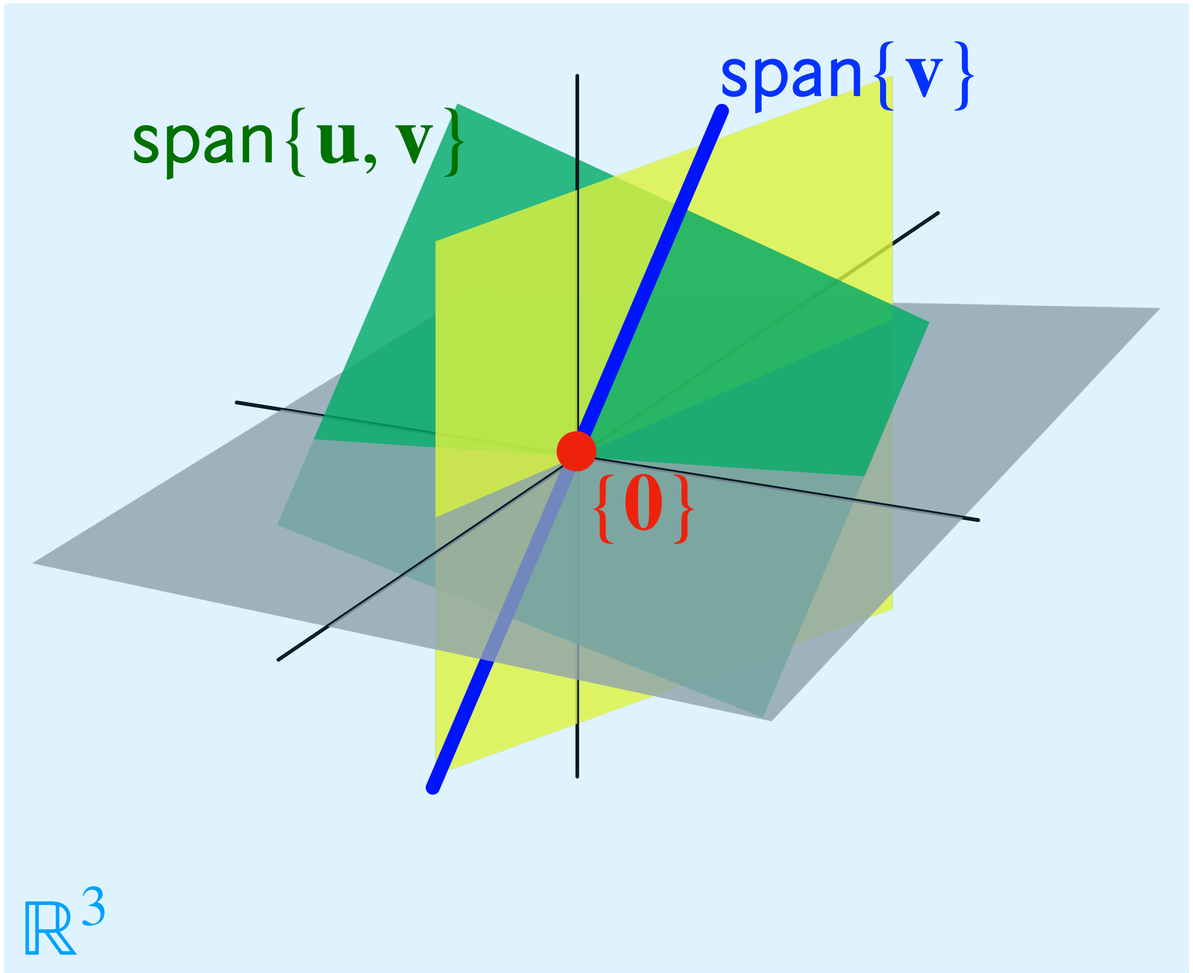


https://commons.wikimedia.org/wiki/File:Linear_subspaces_with_shading.svg

Recall: Subspace in $\mathbb{R}^3$

There are only 4 kinds of subspaces of $\mathbb{R}^3$:

1. 0-dimensional subspace
2. 1-dimensional subspace
3. 2-dimensional subspace
4. 3-dimensional subspace



https://commons.wikimedia.org/wiki/File:Linear_subspaces_with_shading.svg

How does this connect to null space and column space?

Recall: An Observation

The *number* of vectors in the basis we found is the same as the number of *free variables* in a general form solution

$x_1 = 2x_2 + x_4 - 3x_5$

x_2 is free

$x_3 = (-2)x_4 + 2x_5$

x_4 is free

x_5 is free

$$\equiv \quad \begin{bmatrix} s \\ t \\ u \end{bmatrix} \mapsto \begin{bmatrix} 2s + t - 3u \\ s \\ (-2)t + 2u \\ t \\ u \end{bmatrix}$$

Dimension of the Null Space

The **dimension of** $\text{Nul}(A)$ is the number of *free variables* in a general form solution to $A\mathbf{x} = \mathbf{0}$

$x_1 = 2x_2 + x_4 - 3x_5$

x_2 is free

$x_3 = (-2)x_4 + 2x_5$

x_4 is free

x_5 is free

$$\equiv \quad \begin{bmatrix} s \\ t \\ u \end{bmatrix} \mapsto \begin{bmatrix} 2s + t - 3u \\ s \\ (-2)t + 2u \\ t \\ u \end{bmatrix}$$

Recall: An Observation

The *number* of vectors in the Nul(A) basis we found is the same as the number of *free variables* or equivalently the number of *non-pivot columns*

$$[\mathbf{a}_1 \quad \mathbf{a}_2 \quad \mathbf{a}_3 \quad \mathbf{a}_4 \quad \mathbf{a}_5] \sim \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Dimension of the Column Space

The **dimension of** $\text{Col}(A)$ is the number of *basic variable* in our solution, or equivalently the number of *pivot columns* of A

$$[\mathbf{a}_1 \quad \mathbf{a}_2 \quad \mathbf{a}_3 \quad \mathbf{a}_4 \quad \mathbf{a}_5] \sim \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Rank

$= \dim(\mathrm{Col}(A))$

Definition. The **rank** of a matrix A, written rank A, is the dimension of Col A

This is just terminology

Rank-Nullity Theorem

Theorem. For $A \in \mathbb{R}^{m \times n}$, $\text{rank}(A) + \dim(\text{Nul}(A)) = n$

nullity A

$\dim(\text{Col}(A))$

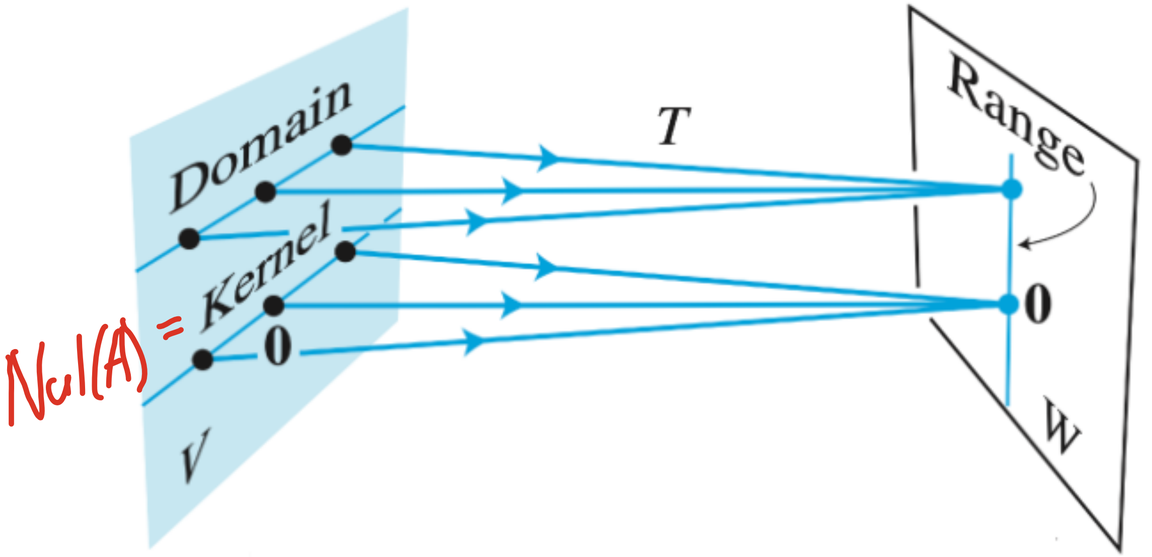
This is incredibly important

Linear Transformations Perspective

If A implements T then

» Col(A) is the same as ran(T), where vectors are "sent" by T

» Nul(A) is the set of vectors that are "zeroed out" by T, also called the **kernel** of T



The Intuition

For $A \in \mathbb{R}^{m \times n}$, $\text{Col}(A)$ *could* be n-dimensional, then

$$\text{rank}(A) + \dim(\text{Nul}(A)) = n + 0 = n$$

But the null space can "consume" some of those dimensions

If a "line's worth of stuff" is pulled into the null space (and mapped to $\mathbf{0}$) then

$$\text{rank}(A) + \dim(\text{Nul}(A)) = (n - 1) + 1 = n$$

The null space "takes away" dimension from the column space

Extending the IMT

Theorem. For an $n \times n$ invertible matrix A, the following are logically equivalent (they must all by true or all by false

- $\text{Col}(A) = \mathbb{R}^n$
- $\dim(\text{Col}(A)) = n$
- $\text{rank}(A) = n$
- $\text{Nul}(A) = \{\mathbf{0}\}$
- $\dim(\text{Nul}(A)) = 0$

workshop (part 3)

Q6

A is $\underset{m}{8} \times \underset{n}{11}$

$$\text{rk } A + \text{nullity } A = \text{rk } A + \dim(\text{Nul}(A)) = n$$

$$\dim(\text{Row } A) + \dim(\text{Nul}(A^T)) = m$$

Q7| A is 9×15 — A^T lin. ind columns



rows are lin. ind. — $\dim \text{Row } A = 9$?

$\text{Row}(A) = \text{span}\{\text{rows}\} \subseteq \mathbb{R}^n$ ↑

Fact: Row A doesn't change under elementary row operations

$$\begin{bmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ 0 & 0 & \cdots & 1 \end{bmatrix} \text{ or } \begin{bmatrix} 1 & * & 0 & * & 0 \\ & & 1 & * & 0 \\ & & & & 1 \end{bmatrix}$$

must have 9 pivot positions $= \dim(\text{Col}(A))$

$\dim(\text{Nul } A) = 15 - 9 = 6$

$$\dim(\text{Row } A) = \dim(\text{Col } A) = \#\text{ of pivot positions in RREF}$$

$$\underbrace{\text{rk } A^T}_{\text{rk } A}$$

$$\dim(\text{Row } A) + \dim(\text{Nul}(A)) = n$$

Later: Row A orthogonal to Nul A as subspaces of $\mathbb{R}^n$